```
!pip install -q groq
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 143.7/143.7 kB 4.4 MB/s eta 0:00:00
```

```
from google.colab import userdata
from groq import Groq
```

```
api_key = userdata.get("Secret_Key")

client = Groq(api_key=api_key)

print("Groq API key loaded successfully!")
```

```
Groq API key loaded successfully!
```

```
response = client.chat.completions.create(
    model="llama-3.1-8b-instant",
    messages=[
        {
            "role": "user",
            "content": "What is Artificial Intelligence?"
        }
    ]
)

print(response.choices[0].message.content)
```

```
Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and lea

AI technology is based on the principle of creating algorithms (sets of instructions) that can process data, recognize patte

1. **Learning**: AI systems can learn from data and improve their performance over time.
2. **Problem-solving**: AI can identify problems, analyze them, and propose solutions.
3. **Decision-making**: AI can make decisions based on data analysis and algorithms.
4. **Pattern recognition**: AI can recognize patterns in data and make predictions.

There are several types of AI, including:

1. **Narrow or Weak AI**: Designed to perform a specific task, such as facial recognition or language translation.
2. **General or Strong AI**: A hypothetical AI that possesses human-like intelligence and can perform any intellectual task.
3. **Superintelligence**: A hypothetical AI that is significantly more intelligent than the best human minds.

The applications of AI are vast and varied, including:

1. **Virtual assistants**: Siri, Alexa, and Google Assistant.
2. **Image and speech recognition**: Facebook's facial recognition software and Google's speech-to-text.
3. **Natural language processing**: Chatbots and language translation software.
4. **Robotics**: Industrial robots and humanoid robots like Sophia.
5. **Healthcare**: AI-powered diagnosis and treatment planning.

While AI has the potential to greatly benefit society, it also raises concerns about job displacement, bias, and accountabil
```

```
system_prompt = """
You are a helpful AI assistant for college students.

Your responsibilities are:

1. Answer questions clearly and accurately.
2. Explain technical concepts in simple language.
3. Give examples whenever useful.
4. Break complex problems into smaller steps.
5. Help with programming and academic questions.
6. Provide Python code when requested.
7. Explain the code in a simple way.
8. If the question is unclear, ask the user for clarification.
9. Do not provide unnecessary information.
10. Keep responses concise, useful, and easy to understand.

Always understand the user's request before providing the answer.
"""
```

```
def ai_assistant(user_query):
```

```
    response = client.chat.completions.create(
        model="llama-3.1-8b-instant",
        messages=[
            {
                "role": "system",
                "content": system_prompt
            },
            {
                "role": "user",
                "content": user_query
            }
        ],
        temperature=0.7
    )

    return response.choices[0].message.content
```

```
question = "Explain Machine Learning in simple words."

answer = ai_assistant(question)

print("AI Assistant:")
print(answer)
```

```
AI Assistant:
Machine Learning (ML) is a way for computers to learn from data without being explicitly programmed. Imagine you're trying t

**Traditional Approach:**

1. You would show the child a picture of a dog and say, "This is a dog."
2. Then, you would show them a picture of a cat and say, "This is a cat."
3. You would repeat this process many times, teaching the child to recognize the differences between dogs and cats.

**Machine Learning Approach:**

1. You would show the child a large collection of pictures, some of dogs and some of cats.
2. The child would look at the pictures and try to figure out the differences between them.
3. Over time, the child would start to recognize patterns and learn to identify dogs and cats independently.

In Machine Learning, computers do the same thing. They're shown large datasets, and through algorithms and statistical model

There are three main types of Machine Learning:

1. **Supervised Learning:** The computer is trained on labeled data, where the correct output is already known.
2. **Unsupervised Learning:** The computer is given unlabeled data, and it must find patterns and relationships on its own.
3. **Reinforcement Learning:** The computer learns through trial and error, receiving rewards or penalties for its actions.

Machine Learning has many applications, including image and speech recognition, natural language processing, and predictive

Do you have any specific questions about Machine Learning?
```

```
question = "What is the difference between AI, Machine Learning and Deep Learning?"

answer = ai_assistant(question)

print("AI Assistant:")
print(answer)
```

```
AI Assistant:
I'd be happy to explain the difference between AI, Machine Learning, and Deep Learning in simple terms.

**Artificial Intelligence (AI):** AI is the broader field that focuses on creating machines that can perform tasks that ty

- Understanding natural language
- Recognizing images and speech
- Making decisions based on data
- Solving problems that require reasoning and learning

AI is like the umbrella term that encompasses various techniques and technologies to achieve these goals.

**Machine Learning (ML):** Machine Learning is a subset of AI that focuses on training algorithms to learn from data and i

Think of Machine Learning like a student who learns from their mistakes and improves their grades over time. The algorithm

**Deep Learning (DL):** Deep Learning is a type of Machine Learning that uses neural networks with multiple layers to anal

Deep Learning is like a very smart student who can recognize patterns and relationships in complex data, such as images, s

To illustrate the hierarchy:

- AI is like a broad field that encompasses various techniques and technologies.
- Machine Learning is a subset of AI that focuses on training algorithms to learn from data.
- Deep Learning is a type of Machine Learning that uses neural networks to analyze and learn from data.

Here's an example in Python to give you an idea of how these concepts might be applied:
```

```
# Example using Scikit-learn (Machine Learning)
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Load the iris dataset
iris = load_iris()
X_train, X_test, y_train, y_test = train_test_split(iris.data, iris.target, test_size=0.2, random_state=42)

# Train a logistic regression model
model = LogisticRegression()
model.fit(X_train, y_train)

# Example using Keras (Deep Learning)
from keras.models import Sequential
from keras.layers import Dense

# Define the neural network architecture
model = Sequential()
model.add(Dense(64, activation='relu', input_shape=(784,)))
model.add(Dense(32, activation='relu'))
model.add(Dense(10, activation='softmax'))

# Compile the model
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

# Train the model
model.fit(X_train, y_train, epochs=10, batch_size=128)
```
Note that this is a simplified example, and actual implementations would require more complexity and nuance.

```
question = "Write a Python program to check whether a number is prime."

answer = ai_assistant(question)

print("AI Assistant:")
print(answer)
```

```
AI Assistant:
**Prime Number Checker in Python**
====================================

Here's a simple Python program that checks if a number is prime:

```python
def is_prime(n):
    """
    Check if a number is prime.

    Args:
        n (int): The number to check.

    Returns:
        bool: True if the number is prime, False otherwise.
    """
    if n <= 1:
        return False
    for i in range(2, int(n ** 0.5) + 1):
        if n % i == 0:
            return False
    return True

# Example usage:
num = int(input("Enter a number: "))
if is_prime(num):
    print(f"{num} is a prime number.")
else:
    print(f"{num} is not a prime number.")
```

How it works:

1. The `is_prime` function takes an integer `n` as input.
2. If `n` is less than or equal to 1, it's not prime, so we return `False`.
3. We only need to check divisibility up to the square root of `n`, so we use a `for` loop to iterate from 2 to the square r
4. Inside the loop, we check if `n` is divisible by the current number `i` by using the modulo operator (`n % i == 0`). If i
5. If we reach the end of the loop without finding a divisor, we return `True`, indicating that `n` is prime.

Example use cases:

* Enter a prime number, such as 23, to see that it's indeed prime.
* Enter a non-prime number, such as 24, to see that it's not prime.
```

```
question = "Explain the concept of Object Oriented Programming and its four main principles."

answer = ai_assistant(question)
```

```
print("AI Assistant:")
print(answer)
```

```
AI Assistant:
**Object Oriented Programming (OOP)**

Object Oriented Programming is a programming paradigm that revolves around the concept of objects and classes. It's a way

**Four Main Principles of OOP:**

1. **Encapsulation**:
Encapsulation is the idea of bundling data and methods that operate on that data within a single unit, called a class. Thi

Example:
```python
class BankAccount:
    def __init__(self, balance):
        self.__balance = balance

    def get_balance(self):
        return self.__balance

    def deposit(self, amount):
        self.__balance += amount
```
In this example, the `BankAccount` class encapsulates the `balance` data and provides public methods to interact with it.

2. **Abstraction**:
Abstraction is the practice of showing only the necessary information to the outside world while hiding the internal imple

Example:
```python
class PaymentGateway:
    def __init__(self):
        self.__api_url = "https://api.example.com"

    def process_payment(self):
        # This method abstracts the actual payment processing
        # and only exposes the necessary information
        pass
```
In this example, the `PaymentGateway` class abstracts the actual payment processing and only exposes the necessary informa

3. **Inheritance**:
Inheritance is the mechanism by which one class can inherit the properties and methods of another class. This helps to pro

Example:
```python
class Vehicle:
    def __init__(self, color):
        self.color = color

class Car(Vehicle):
    def __init__(self, color, num_doors):
        super().__init__(color)
        self.num_doors = num_doors
```
In this example, the `Car` class inherits the `color` attribute from the `Vehicle` class.

4. **Polymorphism**:
```

```
print("====================================")
print("       AI STUDENT ASSISTANT")
print("====================================")
print("Ask your questions.")
print("Type 'exit' to stop the assistant.")
print("====================================")

while True:

    user_input = input("\nYou: ")

    if user_input.lower() == "exit":
        print("\nAI Assistant: Goodbye!")
        break

    answer = ai_assistant(user_input)

    print("\nAI Assistant:")
    print(answer)
```

```
====================================
       AI STUDENT ASSISTANT
====================================
Ask your questions.
Type 'exit' to stop the assistant.
====================================
```

```
You: what do you mean by agentic AI

AI Assistant:
Agentic AI refers to a type of artificial intelligence that can take deliberate actions and make decisions to achieve goals.

In other words, agentic AI is designed to act on its own, rather than simply responding to inputs or following rules. It's l

Agentic AI can be seen in various applications, such as:

1. Autonomous vehicles: Self-driving cars that navigate through traffic and make decisions to avoid obstacles.
2. Personal assistants: Virtual assistants like Siri, Alexa, or Google Assistant that can understand voice commands and take
3. Robotics: Robots that can learn from experience and adapt to new situations.

Agentic AI is different from other types of AI, such as:

1. Reactive AI: Responds to inputs, but doesn't make decisions or take actions.
2. Limited memory AI: Can remember past experiences, but doesn't use that knowledge to make decisions.

To illustrate the difference, consider a simple example:

Reactive AI: A chatbot that responds to "What's the weather like?" with "The weather is sunny." (It's just providing informa

Agentic AI: A self-driving car that decides to take a detour to avoid traffic congestion. (It's taking a deliberate action b

Keep in mind that agentic AI is still a developing field, and there are many challenges and concerns surrounding its develop

Would you like me to elaborate on any specific aspect of agentic AI?

You: exit

AI Assistant: Goodbye!
```

```
conversation_history = [
    {
        "role": "system",
        "content": system_prompt
    }
]
```

```
def memory_assistant(user_query):

    conversation_history.append(
        {
            "role": "user",
            "content": user_query
        }
    )

    response = client.chat.completions.create(
        model="llama-3.1-8b-instant",
        messages=conversation_history,
        temperature=0.7
    )

    answer = response.choices[0].message.content

    conversation_history.append(
        {
            "role": "assistant",
            "content": answer
        }
    )

    return answer
```

```
print("====================================")
print("     MEMORY-BASED AI ASSISTANT")
print("====================================")
print("Type 'exit' to stop.")
print("====================================")

while True:

    user_input = input("\nYou: ")

    if user_input.lower() == "exit":
        print("\nAI Assistant: Goodbye!")
        break

    answer = memory_assistant(user_input)
```

```
    print("\nAI Assistant:")
    print(answer)
```

```
======================================
     MEMORY-BASED AI ASSISTANT
======================================
Type 'exit' to stop.
======================================

You: hi

AI Assistant:
Hello! How can I assist you today? Do you have a specific question about a college course, programming, or something else?

You: tell me about the programming language

AI Assistant:
There are many programming languages out there, each with its own strengths and weaknesses. Here are some popular ones:

1. **Python**: Known for its simplicity, readability, and versatility. It's widely used in web development, data analysis, m
2. **Java**: A popular language for Android app development, web development, and enterprise software development. It's know
3. **JavaScript**: Used for client-side scripting on the web, creating interactive web pages, and developing desktop and mob
4. **C++**: A powerful language for building operating systems, games, and high-performance applications. It's known for its
5. **C#**: A modern language for building Windows applications, games, and enterprise software. It's known for its simplicit

These are just a few examples, and there are many more languages out there. What specific aspect of programming languages wo

Would you like to learn more about:
- Getting started with programming
- Choosing a programming language
- Writing your first program
- Something else?

You: exit

AI Assistant: Goodbye!
```

```
Start coding or generate with AI.
```